

Task 1. Create a Workbench Notebook

1. In the Google Cloud console, from the **Navigation menu** () , select **Vertex AI > Dashboard**.
2. Click **Enable All Recommended APIs**.
3. In the **Navigation menu**, click **Workbench**.

At the top of the Workbench page, ensure you are in the **Instances** view.

4. Click  **Create New**.
5. **Configure the Instance:**
 - **Name:** lab-workbench
 - **Region:** Set the region to us-east4
 - **Zone:** Set the zone to us-east4-b
 - **Advanced Options** (Optional): If needed, click "Advanced Options" for further customization (e.g., machine type, disk size).

New instance



Name *

lab-workbench

Must start with a letter followed by up to 47 lowercase letters, numbers, or hyphens (-) and cannot end with a hyphen

Region *

us-east1 (South Carolina) ▼ ?

Zone *

us-east1-c ▼ ?

Some regions are restricted due to a policy set by your organization. [Learn more](#)

☐ Attach 1 NVIDIA T4 GPU

☒ Enable Dataproc Serverless Interactive Sessions

Enable access to Dataproc Spark kernels

☒ Network in this project

☐ Shared network

Network

default ▼ ?

Subnetwork *

default(10.142.0.0/20) ▼ ?

Instance properties

Machine type	e2-standard-4
Data disk	100 GB Balanced persistent disk
Permission	Compute Engine default service account
Estimated cost ?	\$145.20 monthly, \$0.20 hourly

ADVANCED OPTIONS

CANCEL

CREATE

6. Click **Create**.

This will take a few minutes to create the instance. A green checkmark will appear next to its name when it's ready.

7. Click **Open Jupyterlab** next to the instance name to launch the JupyterLab interface. This will open a new tab in your browser.

View:

INSTANCES

USER-MANAGED NOTEBOOKS

MANAGED NOTEBOOKS

Workbench Instances have JupyterLab 3 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

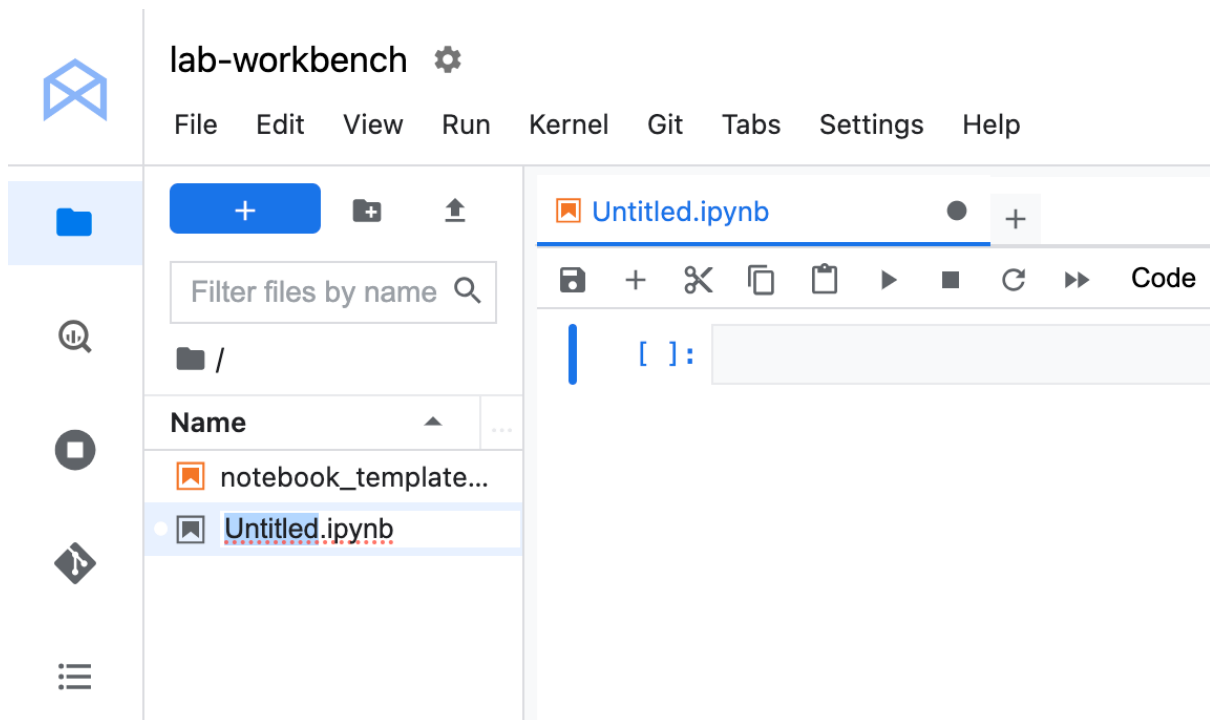
☐	☐	Instance name ↑	Zone	Auto upgrade	Version
☐	☒	lab-workbench	us-east1-c	—	M125

[OPEN JUPYTERLAB](#)

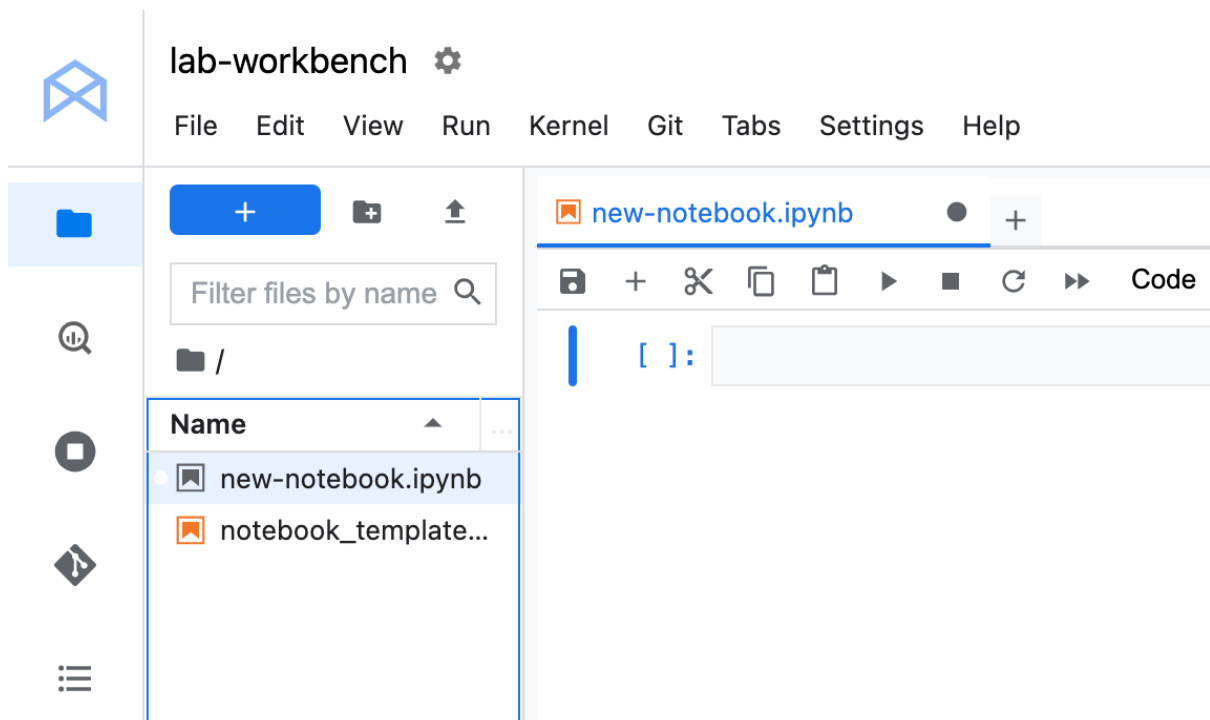
8. Click the **Python 3** icon to launch a new Python notebook.

The screenshot shows the JupyterLab interface. On the left is a sidebar with a file browser showing a folder named 'lab-workbench' containing a file named 'notebook_template...'. The main area is titled 'Launcher' and displays three categories of icons: 'Notebook', 'Console', and 'Other'. Under 'Notebook', there are three icons: 'Python 3 (ipykernel)' (highlighted with a red box), 'PyTorch 1-13 (Local)', and 'TensorFlow 2-11 (Local)'. Under 'Console', there are the same three icons. Under 'Other', there are five icons: 'Terminal', 'Text File', 'Markdown File', 'Python File', and 'Show Contextual Help'.

9. Right-click on the Untitled.ipynb file in the menu bar and select **Rename Notebook** to give it a meaningful name.



Your environment is set up. You are now ready to start working with your Vertex AI Workbench notebook.

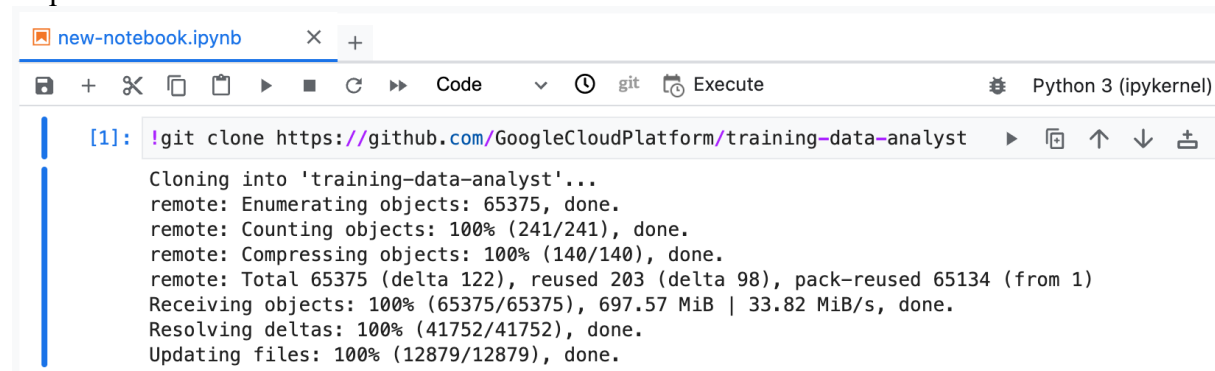


Task 2. Clone a repo within your Vertex AI Notebook instance

The GitHub repo contains both the lab file and solutions files for the course.

1. Copy and run the following code in the first cell of your notebook to clone the training-data-analyst repository.

`!git clone https://github.com/GoogleCloudPlatform/training-data-analyst`
Copied!



The screenshot shows a Vertex AI Notebook interface. At the top, there's a tab labeled 'new-notebook.ipynb'. Below the tab is a toolbar with icons for saving, adding, deleting, and running code, along with a 'Code' dropdown and a 'git' icon. The main area contains a code cell with the command `!git clone https://github.com/GoogleCloudPlatform/training-data-analyst`. The output of the command is displayed below the code, showing the progress of cloning the repository.

```
[1]: !git clone https://github.com/GoogleCloudPlatform/training-data-analyst

Cloning into 'training-data-analyst'...
remote: Enumerating objects: 65375, done.
remote: Counting objects: 100% (241/241), done.
remote: Compressing objects: 100% (140/140), done.
remote: Total 65375 (delta 122), reused 203 (delta 98), pack-reused 65134 (from 1)
Receiving objects: 100% (65375/65375), 697.57 MiB | 33.82 MiB/s, done.
Resolving deltas: 100% (41752/41752), done.
Updating files: 100% (12879/12879), done.
```

2. Confirm that you have cloned the repository. Double-click on the training-data-analyst directory and ensure that you can see its contents.



instance-20240906-103404


File Edit View Run Kernel Git Tabs Settings Help










Filter files by name


 / training-data-analyst /

Name	Last Modified
 blogs	10 minutes ago
 bootcamps	10 minutes ago
 courses	10 minutes ago
 CPB100	10 minutes ago
 datalab	10 minutes ago
 doc	10 minutes ago
 iot	10 minutes ago
 quests	10 minutes ago
 self-paced-labs	10 minutes ago
 CODEOWNERS	10 minutes ago
 CONTRIBUTING.md	10 minutes ago
 learning_rate.ipynb	10 minutes ago
 LICENSE	10 minutes ago

- In the notebook interface, navigate to **training-data-analyst > courses > machine_learning > deepdive2 > launching_into_ml > solutions** and open **workbench_explore_data.ipynb**.
- In the notebook interface, click on **Edit > Clear All Outputs** (click on Edit, then in the drop-down menu, select Clear All Outputs).
- In the **Select Kernel** dialog, choose **Python 3** from the list of available kernels.
- Carefully read through the notebook instructions.



my-new-instance

File Edit View Run Kernel Git Tabs Settings



Filter files by name



/ ... / launching_into_ml / solutions /

Name



Last Modified

 automl_text_classification.ipynb	2 minutes ago
 basic_intro_logistic_regression.ipynb	2 minutes ago
 bigquery.ipynb	2 minutes ago
 decision_trees_and_random_forests.ipynb	2 minutes ago
 explore_data_in_bigquery_with_python.ipynb	2 minutes ago
 explore_data.ipynb	2 minutes ago
 first_model.ipynb	2 minutes ago
 improve_data_quality.ipynb	2 minutes ago
 intro_linear_regression.ipynb	2 minutes ago
 intro_logistic_regression.ipynb	2 minutes ago
 python.BQ_explore_data.ipynb	2 minutes ago
 rapid_prototyping bqml auto ml.ipynb	2 minutes ago
 repeatable_splitting.ipynb	2 minutes ago
 taxi-test.csv	2 minutes ago
 taxi-train.csv	2 minutes ago
 taxi-valid.csv	2 minutes ago
 TrainingWithXGBoostInCMLE.ipynb	2 minutes ago
 workbench_explore_data.ipynb	2 minutes ago

Congratulations!

In this lab you learned how to:

- Create a Workbench Instance Notebook
- Clone a GitHub repository
- Connect to a BigQuery dataset
- Perform statistical analysis on a Pandas Dataframe
- Create Seaborn plots for Exploratory Data Analysis in Python
- Write a SQL query to pick up specific fields from a BigQuery dataset.

End your lab

When you have completed your lab, click **End Lab**. Google Skills removes the resources you've used and cleans the account for you.

You will be given an opportunity to rate the lab experience. Select the applicable number of stars, type a comment, and then click **Submit**.

The number of stars indicates the following:

- 1 star = Very dissatisfied
- 2 stars = Dissatisfied
- 3 stars = Neutral
- 4 stars = Satisfied
- 5 stars = Very satisfied

You can close the dialog box if you don't want to provide feedback.

For feedback, suggestions, or corrections, please use the **Support** tab.

Manual Last Updated November 25, 2024

Lab Last Tested November 25, 2024

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